Peer Review for paper on Network Topology and Manipulative Agents

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The report investigates how network topology affects the influence of manipulative agents in bounded-confidence opinion dynamics. The research question is well defined, and the model (the Hegselmann–Krause model) fits the problem. Section 2 covers the related literature, although the distinction between this project's contribution and the work of Tarantino et al. (2026) could be explained more explicitly. However, I think the report needs to make a stronger distinction between the effects of network topology and network density. I believe that this is especially important because the main conclusion of the paper attributes the observed differences in manipulation effectiveness to topology, even though the experiments do not fully isolate this. Although then later, in Section 5.2, the authors do acknowledge this as a limitation. Yet still, I would perhaps recommend addressing it more directly in the experimental design.

In Section 3, the model is explained through equations and an implementation description, but I believe that in order to fully reproduce the experiments, one would have to make certain assumptions, as the details are not explicitly mentioned. For example, how are the opinions of normal and stubborn agents initialized? Are they sampled uniformly? And are the starting conditions kept the same across experiments? I believe that this is relevant to mention because the distribution of opinions can affect which agents fall within the confidence threshold and how opinions evolve. There is also an inconsistency in the manipulator trajectory, as in Section 3.2 the authors define it as moving from -1 to 1 yet Figures 1 to 3 and Section 5.1 indicate a starting point of approximately -0.6. I believe an improvement to this would specify the function used in the implementation and make it consistent throughout the report. Some equations also need correction, such as Equation 2, which does not express the confidence and connectivity conditions correctly. Additionally, Equation 4 should clarify the orientation of the opinion vector and matrix multiplication.

Another point that needs clarification is the control condition in Section 3.6. The authors state that simulations contain 120 agents (including 10 stubborn agents and either 0 or 10 manipulators). This seems to imply that the number of normal agents changes from 110 to 100 between conditions. Since $\Delta\mu$ is calculated using only normal agents, part of the measured difference could then therefore result from comparing populations with different numbers of normal agents rather than from the actual manipulation itself. I would suggest explaining how agents are selected for each condition, and I believe it would be helpful to explain that the same normal agents are evaluated in both cases. Keeping the network and starting opinions fixed between paired simulations would also help isolate the effect of manipulation.

However, my main concern with the experimental design is how the three topologies are compared. In Section 3.6 each topology is evaluated with its own hyperparameter range, but these parameters do not necessarily produce the same changes in connectivity for each topology. For example, increasing the attachment parameter in a Barabási–Albert network would not necessarily produce the same density as increasing the connection probability in an Erdős–Rényi network. Therefore, the differences in average $\Delta\mu$ may partly reflect the differing densities rather than just

the topology itself. This is recognized in Section 5.2, but I think it deserves more attention because it directly affects the research question. I would suggest perhaps comparing the different networks at matching edge densities (or average degrees) and then examining whether differences in manipulation effectiveness are still present.

I believe that the figures are very helpful in illustrating how opinions evolve, but some of the presentation choices make the results harder to interpret. Figures 2 and 3 have (what I assume to be) a placeholder "HYPERPARAMETER". In addition, Figures 1 to 3 show individual simulation runs, so any observations about cluster formation or the behavior of manipulators should not be generalized from these examples alone. Showing mean trajectories across runs (or including variability around them) could make the findings easier to assess or comment on.

There is also an inconsistency in how the statistical results are interpreted in Section 4. The authors state that random and Watts–Strogatz networks differ significantly, while Barabási–Albert and Watts–Strogatz networks do not. However, Table 2 shows the opposite, and Section 5.1 correctly interprets this. I would recommend correcting this. Additionally, Section 3.6 mentions averaging over 10 runs, while Section 4 refers to 100 seeds. Clarifying how many runs were used and how the statistical tests were performed would be helpful for reproducibility.

The conclusion in Section 6 makes a distinction between average and maximum manipulation effects, which I think helpful and definitely worth including. However, the claim that Barabási–Albert networks are more susceptible to manipulation should be framed as a finding specifically under the parameter settings that were tested, rather than as a general property of that topology.

Overall, I would suggest focusing on the reproducibility of the model and the comparability of the network experiments. In particular, controlling for network density and explaining how the manipulation effect is isolated would help to establish whether the observed differences can be attributed to topology (which is ultimately the question the project is aiming to answer).